Decomposition of Monolith Applications Into Microservices Architectures: A Systematic Review

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Paper Information: Key Paper, Survey, Systematic Literature Review (SLR)

Rating: ⭐ ⭐ ⭐ ⭐ ⭐

Own Keywords: Background, Microservices, Migration, Monolith, Monolith Decomposition

Reading Progress: First Pass

Introduction (What is the Paper about?)

- **Aim: give an overview about the field of *Monolith Decomposition***
  - includes different, primary phases of monolith decomposition
  - identifying research gaps
  - outlining future directions
- **Comprehensive and interrelated picture of various decomposition techniques, benchmarks and metrics**
  - outlining differing migration processes
  - identify existing tooling and methods
  - identify current metrics, datasets and benchmarks
  - investigate contributions seeking to define migration road maps
  - problems and challenges during migration

Methods

- ***Systematic Literature Review (SLR)*** protocol and *snowballing* method
  - identify 5,022 search results from major computer science literature databases
  - applies four refinement steps concluding to 33 papers
  - after applying snowballing research finally ends up with 35 papers
- **Examination and Evaluation using follow-up research** after the original paper was made public

Scientific Benefit

- **Introduced the *Monolith to Microservices Decomposition Framework (M2MDF)*** which identifies the mayor phases and key elements of decomposition
  - provides a comprehensive map for researchers and practitioners
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- **Detailed Analysis of existing Decomposition approaches, tools and methods**
  - monolith decomposition remains at an early stage and remains a complicated and expensive task
    - lack of integrated, comprehensive data collection and analysis methods with respect to crucial monolith aspects
    - lack of comparison between various decomposition methods
    - insufficient tool support, standardized metrics and datasets

    - absence of methods combining static, dynamic and evolutionary data
- **Identification of used metrics and datasets for Evaluation**
- **can assist practitioners seeking to understand the dimensions of monolith decomposition** and current capabilities
    - findings valuable for researches looking to identify areas of further extension

Challenges / Future Work

Limitations

Paper Outline and Read Progress

1. [X] Introduction
2. [X] Background and Related Work
3. [X] SLR Methodology
4. [X] Presentation of the *M2MDF*
    - Detailed Results of the SLR
5. [ ] Existing Gaps in the field
6. [ ] Threats to validity
7. [ ] Summary, Discussion
    - Specific implications for practitioners and researchers
    - Inclusion of post-review reflection evaluating M2MDF examining follow-up literature
8. [ ] Conclusion and future research directions

Own Comments

(+) Key Paper